Optimization for Data Science

M2 MIAGE / MIAGE ID Apprentissage

Project – 2026/2027

Optimization with imperfect information

Can we learn without exact gradients?

Assignment. Submit one report (English or French) and one completed, executable Python notebook per group of one or two students. The report must answer the theoretical questions, state your predictions before discussing the results, justify your experimental choices, and explain a counter-experiment. Include your group code, random seed, software versions, plots with labeled axes, and sufficient information to reproduce each figure. An AI assistant may be used, but the group is responsible for the correctness of all claims and code. No GPU is required. Expected effort: roughly 10–15 hours per student. *The submission deadline and submission platform will be announced by the teaching team.*

Introduction

Optimization for machine learning usually assumes access to a gradient. In practice, gradients can be unavailable (a black-box model), expensive to compute, or noisy. The choice of an optimizer should therefore reflect not only its convergence in iterations, but also the information used by each iteration.

We study a small binary classification task using a regularized logistic loss. Four methods will be compared: full gradient descent (GD), mini-batch stochastic gradient (SGD), coordinatewise central finite differences (CFD), and simultaneous perturbation gradient estimates (SPSA). GD and SGD have access to derivative information; CFD and SPSA only query values of the objective. Our primary question is: *What accuracy can we achieve with a fixed computational budget, and what do we lose when gradients are not available?*

This is a teaching-scale investigation inspired by research on derivative-free optimization and computationally efficient learning; it is not an implementation of a particular research paper. Relevant material from the course notes includes Chapters 2 (gradient methods), 3 (stochastic gradient and computational trade-offs), and 4 (regularization).

1. A one-dimensional noisy oracle

Let $a \in \mathbb{R}$, $q(w) = \frac{1}{2}(w-a)^2$, and suppose that a query at w returns

$$Y(w) = q(w) + \xi, \qquad \mathbb{E}[\xi] = 0, \quad \mathrm{Var}(\xi) = \tau^2, \tag{1}$$

where noises from *different calls* are independent. A forward-difference estimate is

$$D_h^+(w) = \frac{Y(w+h) - Y(w)}{h}, \qquad h > 0.$$

Question 1. (a) Compute $q'(w)$, $\mathbb{E}[D_h^+(w)]$, its bias, and $\mathrm{Var}(D_h^+(w))$. (b) Deduce the mean-squared error $\mathbb{E}[(D_h^+(w) - q'(w))^2]$. (c) For $\tau > 0$, find the value of h minimizing this error. What changes if $\tau = 0$? Explain the competing effects of a small and a large h.

1. A one-dimensional noisy oracle (continued)

A central-difference estimate uses two *new, independent* oracle calls:

$$D_h^0(w) = \frac{Y(w+h) - Y(w-h)}{2h}.$$

Question 2. (a) Compute its bias, variance, and mean-squared error for the quadratic q. Does this special example suggest that arbitrarily large h is always desirable? (b) Consider instead $q_\gamma(w) = \frac{1}{2}(w-a)^2 + \frac{\gamma}{4}(w-a)^4$ with $\gamma > 0$. Compute the noiseless bias of $D_h^0(w)$ relative to $q_\gamma'(w)$. Explain how this example changes your answer to (a). (c) *Before running the notebook*, predict the shapes of the error-versus-h curves for forward and central differences, both with and without noise. Record your prediction in writing.

2. The learning problem

The data consist of a matrix $X \in \mathbb{R}^{n\times d}$ with rows $x_i^\top$ and labels $y_i \in \{-1, +1\}$. For $\lambda > 0$ we minimize

$$f(w) = \frac{1}{n}\sum_{i=1}^{n} \log(1 + \exp(-y_i x_i^\top w)) + \frac{\lambda}{2}\|w\|^2, \qquad w \in \mathbb{R}^d. \tag{2}$$

The notebook provides a seeded synthetic-data generator, separate training and test sets, a *clean* evaluation function, and a *noisy value oracle* $F(w) = f(w) + \varepsilon$ with independent zero-mean perturbations of standard deviation σ. The clean objective is used *only for plotting and assessment* of an optimizer's progress; methods declared black-box must not access its gradient.

Write $s_i(w) = 1/(1 + \exp(y_i x_i^\top w))$.

Question 3. (a) Derive $\nabla f(w)$ and $\nabla^2 f(w)$. (b) Show that f is λ-strongly convex and that ∇f is L-Lipschitz for

$$L = \frac{\|X\|_2^2}{4n} + \lambda.$$

State which result of the course justifies using a constant GD stepsize $1/L$. (c) Give one reason why a small training loss need not imply the best test classification accuracy. *Hint:* $s_i(1 - s_i) \le 1/4$.

3. Four information regimes

At iteration k, consider the update $w_{k+1} = w_k - \alpha_k g_k$ with the following choices:

$$\begin{aligned}
\text{GD: } & g_k = \nabla f(w_k), \\
\text{SGD: } & g_k = \frac{1}{b}\sum_{i\in B_k} \nabla \ell_i(w_k) + \lambda w_k, \\
\text{CFD: } & [g_k]_j = \frac{F(w_k + he_j) - F(w_k - he_j)}{2h}, \quad j = 1, \dots, d, \\
\text{SPSA: } & g_k = \frac{F(w_k + hu_k) - F(w_k - hu_k)}{2h}\, u_k,
\end{aligned}$$

where $\ell_i(w) = \log(1 + \exp(-y_i x_i^\top w))$, B_k is a randomly sampled mini-batch of size b, $u_k \in \{-1, +1\}^d$ has independent Rademacher entries, and *each* F call has fresh noise.

Question 4. Verify that $\mathbb{E}[u_k u_k^\top] = I$. Using a first-order Taylor expansion, explain why SPSA approximates $\nabla f(w)$ when h is small. Is it necessarily an *exactly unbiased* estimator for the logistic objective at fixed $h > 0$? Why not?

4. Implementation and experimental protocol

Implementation 1. Complete the functions marked `TODO` in the accompanying notebook: (i) regularized logistic objective and gradient; (ii) mini-batch gradient; (iii) CFD using the noisy *value-only* oracle; (iv) SPSA using the same oracle; and (v) the update/recording loop for the four methods. Check dimensions, numerical stability, and reproducibility. Do not call automatic differentiation or a ready-made optimizer for the methods under study; NumPy, SciPy, and scikit-learn may be used for validation and data generation.

Data and personalization. Use the provided deterministic generator with $n = 400$, $d \in \{8, 24\}$ and $\lambda = 0.10$. Fix an integer group key in $\{0, \dots, 9999\}$, report it, and use the corresponding seed in the notebook. The key affects the generated features and labels, not the mathematical questions. The generator uses nested datasets: the first eight features and the labels are identical for $d = 8$ and $d = 24$; the additional features are nuisance variables. The *same* training set, start point, and independent random-number streams must be used in all comparisons for a given group.

Work budget. Define one *sample-term evaluation* as evaluating either a single-example loss contribution or its gradient contribution. For n training samples, the nominal work of one step is:

Method	GD	SGD (batch b)	CFD	SPSA
Nominal sample-term evaluations	n	b	$2dn$	$2n$

This is an *operation-count proxy*, not a claim that all four methods take equal wall-clock time per sample term. Also record elapsed CPU/wall time separately. Clean diagnostic evaluations and test-set measurements must be logged separately and *must not be used by the black-box update rules.*

Implementation 2. At one fixed point w (provided in the notebook), estimate the mean squared gradient error $\|g - \nabla f(w)\|^2$ by independent repetitions of CFD and SPSA, for several h values and both $\sigma = 0$ and $\sigma > 0$. Report the average and dispersion. The exact gradient is allowed *in this diagnostic experiment only*, never in CFD or SPSA update code.

Question 5. Explain which aspects of your plots agree with Questions 1–2, which depend on the dimension, and which are not predicted by the one-dimensional quadratic analysis. Justify how you would choose h for the main experiment. The smallest tested h is not automatically the best.

Implementation 3. Run GD, SGD, CFD, and SPSA from the same $w_0 = 0$ on the same dataset and with a common nominal budget (the notebook provides a conservative default). Compare training objective gap to a reference solution, test accuracy, nominal budget, and measured wall time. State clearly how you chose or tuned α_k, b, and h. Do not tune hyperparameters using the test set.

5. Prediction, diagnosis, and counter-experiment

Question 6. Pre-register two predictions. Before looking at the optimization curves, write down: (i) which method you expect to reach the smallest training-objective gap *per nominal sample-term budget*; (ii) how the ranking should change when d increases from 8 to 24 or when oracle noise σ increases. Justify each prediction in no more than six sentences. Put these predictions before your experimental plots in the submitted report.

Question 7. Evidence and diagnosis. Use at least three independent random runs for stochastic algorithms. Compare methods on (a) clean training-objective gap versus nominal budget, (b) clean training-objective gap versus measured time, and (c) test accuracy at the end of training. Give uncertainty or variability information rather than relying on one run. Discuss at least one observation that contradicts, qualifies, or is not explained by your predictions. If no prediction is contradicted, identify a limitation of the evidence and formulate an alternative explanation you have not ruled out.

Question 8. Design your own counter-experiment. Choose *one* claim from your analysis that could be wrong. Change exactly one controlled factor (e.g., dimension, σ, finite-difference scale, mini-batch size, or budget), with all other settings fixed. State the claim, the competing explanation, your experimental change, the resulting evidence, and whether your original conclusion survives. A successful counter-experiment does not need to overturn the claim; it must be able, in principle, to falsify it.

6. Deliverables and assessment

Your report should contain: (1) answers to Questions 1–4; (2) your *predictions written before the plots*; (3) a compact table of the experimental settings; (4) clearly labeled figures and a short critical discussion answering Questions 5–8; (5) the group key and full reproducibility information; and (6) an individual paragraph from each participant explaining one concrete experimental decision, one failure mode, or one revised interpretation. The paragraphs should concern the work, not personal circumstances. Include the completed notebook and a short statement of any AI tools used and what they were used for.

Indicative grading. Mathematical reasoning 20%; correct and reproducible implementations 20%; experiments, uncertainty, and counter-experiment 35%; interpretation and justification 15%; short individual verification 10%. The last component may take the form of a brief individual discussion with the teaching team.

Scope and good practice. The point is not to prove that a derivative-free method always wins, nor that a method with better iteration complexity is automatically cheaper in practice. Carefully distinguish the exact *gradient-oracle* access available to GD/SGD, the *function-value* access available to CFD/SPSA, the nominal operation-count proxy, and actual running time. Your results depend on the chosen data, parameter ranges, randomness, and hyperparameters. Do not turn an observation from one configuration into a universal statement.

Course alignment: Chapters 1–3 of *PolyODS_2627*; regularization in Chapter 4. The finite-difference and SPSA estimators are introduced within this project and do not require an external prerequisite.